Qualifying Exam in Microeconomics
Arizona State University
August 2024

There are four questions and you have four hours to complete the exam. If you believe that a question is in error, please explain the error and continue working.

1. A pure exchange economy has two goods indexed by $\ell = 1, 2$, and two agents indexed by $i = 1, 2$ with the same the utility function

$$U_i(x_{i1}, x_{i2}) = \sqrt{x_{i1}} + \sqrt{x_{i2}}.$$

(Note that $x_{i\ell}$ refers to the consumption of good ℓ by agent i; the first subscript indicates the agent and the second subscript indicates the good.) Endowments are $\omega_1 = (2, 0)$ and $\omega_2 = (0, 2)$ and consumption sets are the positive orthant.

(a) Find the aggregate excess demand function $z(p)$ for this economy.

(b) Use the aggregate excess demand function to find a Walrasian equilibrium price.

(c) For an arbitrary aggregate excess demand function $z(p)$ defined for $p \in \mathbb{R}^2_{++}$, give conditions which taken together guarantee that a Walrasian equilibrium exists.

Confirm that the aggregate excess demand function in part (a) satisfies your conditions.

(d) Give an example of an exchange economy with two goods and two consumers for which no Walrasian equilibrium exists. Explain carefully why no Walrasian equilibrium exists.

2. The inverse demand for a good is $\tilde{p}(Q) = \max\{0, a - Q\}$, where Q is aggregate consumption. There are two firms that produce the good, each with a constant-returns technology with unit cost of k, where $a > k > 0$.

   The timing is that firm 1 first chooses an output $q_1 \geqslant 0$. After observing firm 1's output, firm 2 chooses its output $q_2 \geqslant 0$. Firm i's preferences over outcomes in this extensive form game are represented by profit; and, for $i = 1, 2$, firm i's profit of course equals (here $Q = q_1 + q_2$)

$$U_i(q_1, q_2) = \tilde{p}(Q)q_i - kq_i.$$

   (a) (i) Describe the strategy set S_i of each firm in the extensive-form game. (ii) Let u_i be firm i's utility representation over strategy profiles, $S_1 \times S_2$, in the *strategic*-form representation of the game. What does u_i equal for $i = 1, 2$? (iii) Define the notion of a *Nash equilibrium* for this two-player game (using just the "$u_i(s_1, s_2)$, S_i", notation).

   (b) Consider these two questions about *weakly dominant* and *weakly dominated* strategies for firm 2.

      i. Does firm 2 have a *weakly dominant strategy*?[1] If so, identify it and verify that it is weakly dominant; if not, prove that firm 2 does not have such a strategy.

      ii. If you answered *yes* to *i*, then is the weakly dominant strategy you identified part of *every* Nash equilibrium of this game? Prove it or give a counterexample.
      If you answered *no* to *i*, then is there a *Nash equilibrium* in which firm 2 uses a *weakly dominated* strategy?[2] Explain why or why not.

   (c) Is there a Nash equilibrium of the game that is not subgame perfect? If so, identify it and explain why it is a Nash equilibrium and why it isn't subgame perfect; if not, explain why every Nash equilibrium is subgame perfect.

   (d) Derive a *subgame-perfect* Nash equilibrium of this game, *carefully* justifying your steps.

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[1]Let S_2 be firm 2's (pure) strategies. A strategy $s_2'' \in S_2$ *weakly dominates* $s_2' \in S_2$ if $u_2(s_1, s_2'') \geqslant u_2(s_1, s_2')$ for every $s_1 \in S_1$, and the inequality is strict for at least one $s_1 \in S_1$. Strategy $s_2^* \in S_2$ is *weakly dominant* if it weakly dominates every $s_2 \in S_2$, $s_2 \neq s_2^*$.

[2]Strategy $\tilde{s}_2 \in S_2$ is *weakly dominated* if some $s_2 \in S_2$ weakly dominates it.

3. There are two types of agents indexed by $i = 1, 2$ with utility functions

$$u_1(w_1, e_1) = 100 + 10w_1 - 4e_1^2$$
$$u_2(w_2, e_2) = 100 + 10w_2 - 2e_2^2$$

where $w_i \in \mathbb{R}$ is the salary paid to agent i and $e_i \in [0, 12]$ is the education level chosen by agent i.

The productivity q_i of a type-i agent is determined by agent i's education as follows

$$q_1(e) = 8 + 2e_1$$
$$q_2(e) = 8 + 2.5e_2$$

(a) Derive the equations for the indifference curves of both agents that go through the point $w_i = 10$, $e_i = 2$.

(b) Suppose for this question that employers can identify an agent's type; that is to say, employers can tell whether the agent is $i = 1$ or $i = 2$. Employers offer to pay to each individual i an amount w_i equal to i's actual productivity. For each i calculate the e_i that the agent will choose, the w_i that the agent will be offered, and the corresponding utility level for i.

(c) For the following questions, assume employers cannot tell whether an agent is type $i = 1$ or $i = 2$, but employers can observe their level of education. Suppose that employers offer to pay new hires according to the following rule,

$$w(e) = \begin{cases} 8 + 2e & \text{if } e < e^* \\ 8 + 2.5e & \text{otherwise.} \end{cases}$$

where $w(e)$ is salary paid as a function of the education level observed and the threshold e^* specified in the contract.

For each of the following thresholds, either calculate a separating equilibrium and show that it is an equilibrium or demonstrate that a separating equilibrium does not exist. Assume that in a separating equilibrium, all agents must be paid their actual productivity. (This assumption may be justified by assuming competitive labor market.)

i. $e^* = 4$;

ii. $e* = 9$;

(d) Suppose now that employers offer to pay new hires according to the following rule,

$$w(e) = \begin{cases} q_1(e) & \text{if } e < 4 \\ \mu q_1(e) + (1 - \mu) q_2(e) & \text{otherwise.} \end{cases}$$

where μ is the proportion of type-1 agents in the population and $1 - \mu$ is the proportion type-2 agents in the population.

i. For $\mu \in (0, 1)$, calculate the optimal e for each type.

ii. Is there $\mu*$ at which there is a pooling equilibrium? (In a pooling equilibrium, it is not required that each type be paid their true productivity; at a pooling equilibrium they are all paid the same amount.)

4. A consumer lives in an economy with a single physical good and two states of the world. The single physical good can be consumed in any nonnegative quantity in each state, so the consumption set is $X = \mathbb{R}^2_+$. The consumer's preferences are represented by a *differentiable*, *locally nonsatiated* (LNS), *strictly quasiconcave* function $U : X \to \mathbb{R}$. The consumer's demand at price-wealth pair (p, w) is $d(p, w)$. The consumer's wealth is the value of its consumption endowment, $p \cdot \omega$, where $\omega = (\omega_1, \omega_2)$ with

$$\omega_2 > \omega_1 \geqslant 0.$$

For $t = 0, 1$, let $p^t \in \mathbb{R}^2_{++}$ with state-1 consumption relatively cheaper at p^1:

$$\frac{p_1^0}{p_2^0} > \frac{p_1^1}{p_2^1};$$

and let $x^0 = d(p^0,\, p^0 \cdot \omega)$. Suppose that the consumer buys more of the good in state 1 at $p = p^0$:

$$x_1^0 > x_2^0 \geqslant 0.$$

The question here is this: what can be said about the consumer's choice at the price $p = p^1$? *In particular, does the consumer continue to choose more of the good in state 1?*

(a) (i) Write down what it means for the demand d to satisfy the *weak axiom of revealed preference* (WARP) for a pair of prices, (p^0, p^1). Suppose that $x^1 = d(p^1, p^1 \cdot \omega)$.
(ii) Is it possible that $x_1^1 \leqslant x_2^1$? A picture might help, remembering that $\omega_2 > \omega_1$. (Use the fact that, when there are only two goods, if a pair of choices satisfies budget balance and WARP, then there is a LNS and strictly quasiconcave utility function that rationalizes those choices.)

For the rest of the problem specialize to *subjective expected utility* preferences:

$$U(x) = \pi_1 u(x_1) + \pi_2 u(x_2),$$

where $u : \mathbb{R}_+ \to \mathbb{R}$ is the consumer's vN-M utility function and $\pi_s > 0$ is the consumer's subjective probability that the state is s. (Of course $\pi_1 + \pi_2 = 1$.)

Since U is strictly quasiconcave, it follows that u is strictly concave; and since U is also LNS, it follows that $u' > 0$, two facts about u that you should freely use.

(b) Write down the Kuhn-Tucker conditions for the consumer's problem.

(c) Suppose that $x^1 = d(p^1, p^1 \cdot \omega)$. Is it possible that $x_1^1 \leqslant x_2^1$? Explain. [We suggest that you use part (b).]

(d) Using your answers to parts (a) and (c), summarize in words a "testable implication" of subjective expected utility (for risk averse and locally-nonsatiatied preferences) that can distinguish it from other models of choice.